

6. Adapting SCADA for IP (5 Marks)

Adapting SCADA for IP means modifying traditional SCADA systems to operate over IP-based networks instead of serial communication.

Traditional SCADA

- Used serial communication such as RS-232 and RS-485.
- Used proprietary communication protocols.

SCADA over IP

- Uses Ethernet and TCP/IP networks for communication.
- Allows communication between MTU and RTU over IP networks.
- **Protocols Used:** Modbus TCP, IEC 60870-5-104.

Advantages

- Cost-effective by using existing IP infrastructure.
- Scalable and flexible.
- Enables remote monitoring and control.
- Easy integration with IoT and cloud systems.

Disadvantages

- Increased cybersecurity risks.
- Possible network latency and packet loss.
- Requires strong security mechanisms such as VPN, encryption, and firewalls.

5. Generic Web-Based Protocol (5 Marks)

Generic web-based protocols allow IoT devices to communicate with web servers and cloud platforms using standard internet technologies.

Features

- Based on web technologies such as HTTP.
- Supports communication between clients and servers.
- Enables data exchange in distributed systems.

Examples of Web-Based Protocols

- **1. HTTP (HyperText Transfer Protocol)**
 - Request-response protocol used between client and server.
 - Widely used for websites and REST APIs.
- **2. HTTPS**
 - Secure version of HTTP.
 - Uses SSL/TLS encryption to ensure confidentiality and integrity.
- **3. WebSocket**
 - Provides full-duplex communication between client and server.
 - Suitable for real-time applications.

Advantages

- Platform independent.
- Easy integration with cloud services.
- Uses existing web infrastructure.

4. Data Analytics vs Business Benefits (5 Marks)

Data Analytics

Data analytics is the process of collecting, cleaning, and analyzing data to obtain meaningful insights.

- **Types of Data Analytics:**
 - Descriptive Analytics – Describes what happened.
 - Diagnostic Analytics – Explains why it happened.
 - Predictive Analytics – Predicts what is likely to happen.
 - Prescriptive Analytics – Suggests actions to take.
- Data analytics involves programming, database management, and data visualization.

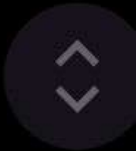
Business Benefits

Business benefits are the measurable improvements achieved by applying data analytics.

- Improves organizational growth and performance.
- Helps in better decision making and strategic planning.
- Increases efficiency and productivity.
- Measured in terms of revenue, cost savings, and ROI.

Difference

Parameter	Data Analytics	Business Benefits
Nature	Technical process	Organizational outcome
Focus	Data processing and insights	Profit and growth
Role	Tool or method	Result or impact
Measurement	Accuracy and reports	Revenue and ROI



3. Short Note on SCADA (5 Marks)

SCADA (Supervisory Control and Data Acquisition) is a system used to monitor and control industrial processes.

Features

- Provides centralized monitoring and control.
- Collects real-time data from sensors and field devices.
- Used in industries such as power plants, manufacturing, and water systems.

Components of SCADA

- **1. Sensors and Actuators**
 - Sensors measure parameters like temperature, pressure, voltage, and current.
 - Actuators perform actions based on control commands.
- **2. Programmable Logic Controllers (PLC)**
 - Connected to sensors and actuators.
 - Collect data from sensors and control equipment.
- **3. Remote Terminal Units (RTU)**
 - Collect data from sensors over large geographical areas.
 - Send the data to SCADA servers.
- **4. SCADA Network**
 - Communication network connecting PLCs, RTUs, and servers.
 - Uses fibre, cable, telephone lines, or satellites.
- **5. Human Machine Interface (HMI)**
 - Interface used by operators to monitor and control systems.

2. IoT Protocols (5 Marks)

IoT protocols enable communication between IoT devices, networks, and cloud platforms.

1. Link Layer Protocols

These protocols manage communication between devices.

- **IEEE 802.3 (Ethernet):** Used for wired communication and widely used in networks.
- **IEEE 802.11 (WLAN):** Wireless communication technology commonly used in IoT systems.
- **IEEE 802.15.4 (LR-WPAN):** Used for low-power wireless personal area networks with low data rate and low cost.
- **IEEE 802.16 (WiMAX):** Developed for broadband wireless communication.
- **Cellular Networks (2G / 3G / 4G LTE / 5G):** Used for connecting IoT devices over long distances.

2. Network Layer Protocols

- **IPv4:** Uses 32-bit addressing and supports approximately 2^{32} addresses.
- **IPv6:** Uses 128-bit addressing and provides a very large address space.
- **6LoWPAN:** Allows IPv6 communication over low-power wireless networks.

3. Transport Layer Protocols

- **TCP:** Reliable and connection-oriented communication.
- **UDP:** Faster, connectionless communication for real-time applications.

4. Application Layer Protocols

- **HTTP / HTTPS:** Used for web communication and REST APIs.
- **CoAP:** Designed for constrained IoT devices.
- **MQTT:** Lightweight publish-subscribe messaging protocol.
- **XMPP:** XML-based messaging protocol.
- **DDS:** High-performance real-time data exchange.
- **AMQP:** Message-oriented middleware protocol.

1. Communication Network Layer (5 Marks)

The Communication Network Layer in IoT connects physical devices such as sensors and actuators to external networks so that data can be transmitted and accessed remotely.

Functions

- Connects IoT devices to networks and the internet.
- Enables communication between sensors, gateways, and cloud servers.
- Uses wired or wireless communication technologies.

Sublayers of Communication Network Layer

- **1. Access Network Sublayer**
 - Connects physical IoT devices to the network.
 - Uses wireless or wired technologies.
 - Examples: Wi-Fi, ZigBee, Bluetooth.
- **2. Gateway and Backbone Network Sublayer**
 - IoT gateways transfer sensor data to the cloud or IoT servers.
 - Gateways process and filter data before forwarding it.
 - Routers forward data from one network to another.
- **3. Network Transport Sublayer**
 - Corresponds to Layer 3 and Layer 4 of the OSI model.
 - Uses TCP/IP and UDP protocols for communication.
- **4. IoT Network Management Sublayer**
 - Uses higher-level protocols for data exchange.
 - Examples: MQTT, CoAP, AMQP.